

The Practice of Confidence Intervals

STAT 200 – Introductory Statistics

Torin Quinlivan

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Today's Objective

After today's lecture, you should:

- State what a confidence interval concerns.
- Calculate confidence intervals using [R](#).
- Understand the analysis procedure for:
 - the value of one population mean.
 - the value of one population proportion.
 - the value of one population variance.
- Understand the analysis procedure for:
 - the difference between two population means (independent).
 - the difference between two population means (dependent).
 - the difference between two population proportions.
 - the ratio of two population variances.

One-Parameter Procedures: μ

Parametric Procedure: Student's t -test

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from NORMAL distribution
 - Requirement test: Shapiro-Wilk test
 - `shapiroTest(x)`
- R function: `t.test(x)`

One-Parameter Procedures: μ (and $\tilde{\mu}$)

Non-Parametric Procedure #1: Wilcoxon procedure

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from symmetric distribution
 - Requirement test: Hildebrand Rule
 - `hildebrand.rule(x)`
- R function: `wilcox.test(x, conf.int=TRUE)`

One-Parameter Procedures: μ

Non-Parametric Procedure #2: Non-parametric bootstrap

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Nothing!
- R code:

```
st = numeric()

for (i in 1:1e4) {
  x = sample(y, replace=TRUE)
  st[i] = mean(x)
}

quantile(st, c(0.025, 0.975))
```

One-Parameter Procedures: p

Parametric Procedure: Binomial procedure

- Graphic: Binomial plot (`binom.plot(x, n)`)
- Requires: Data generated from BINOMIAL distribution
- R function: `binom.test(x, n)`

Note: This is *not* the procedure Hawkes covers. They use something close to what we refer to as the Wald test. The function to perform the Wald test in R is `wald.test`.

One-Parameter Procedures: p

Non-Parametric Procedure: Non-parametric bootstrap

- Graphic: Binomial plot (`binom.plot(x, n)`)
- Requires: Nothing
- R code:

```
st = numeric()

for (i in 1:1e4) {
  x = sample(y, replace=TRUE)
  st[i] = mean(x)
}

quantile(st, c(0.025, 0.975))
```

This code assumes the data, `y`, consist of 0 and 1 values.

One-Parameter Procedures: σ^2

Parametric Procedure: Chi-Square (χ^2) procedure

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Data generated from NORMAL process
 - Requirement test: Shapiro-Wilk test
 - `shapiroTest(x)`
- R function: `onevar.test(x)`

One-Parameter Procedures: σ^2

Non-Parametric Procedure: Non-parametric bootstrap

- Graphic: box-and-whiskers plot (`boxplot(x)`)
- Requires: Nothing
- R code:

```
st = numeric()

for (i in 1:1e4) {
  x = sample(y, replace=TRUE)
  st[i] = var(x)
}

quantile(st, c(0.025, 0.975))
```

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Independent)

Parametric Procedure: two-sample t -procedure

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Data generated from NORMAL distribution – in *each* population
 - Requirement test: Shapiro-Wilk test
 - `shapiroTest(x ~ g)`
- R function: `t.test(x ~ g)`
- R function: `t.test(x1, x2)`

Note: This is not the same test as Hawkes uses. As such you should not use this to do your HLS homework.

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Independent)

Non-Parametric Procedure: Mann-Whitney procedure

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Beyond the scope of this course
- R function: `wilcox.test(x ~ g, conf.int=TRUE)`
- R function: `wilcox.test(x1, x2, conf.int=TRUE)`

Two-Parameter Procedures: $\mu_1 - \mu_2$ (Dependent)

Parametric Procedure: two-sample dependent procedure

- This is just a one-sample procedure if you focus on the differences, $d = \mu_1 - \mu_2$, instead of the measurements.
- As such, refer to the one-sample mean procedure.
- This should help to illustrate the need to understand the data and what you want to understand.

Two-Parameter Procedures: $p_1 - p_2$

Parametric Procedure: Proportions procedure

- Graphic: Binomial plot (`binom.plot(x=c(x1, x2), n=c(n1, n2))`)
- Requires: Expected number of successes is at least 5 in each group
- R function: `prop.test(x=c(x1, x2), n=c(n1, n2))`

Note: This is *not* the procedure Hawkes covers. They use something close to this, but this procedure makes adjustments for the fact that the BINOMIAL distribution is discrete and the NORMAL distribution is not. As such, you should use the Wald test for this type of Hawkes problem, too. (R function: `wald.test`)

Two-Parameter Procedures: σ_1^2/σ_2^2

Parametric Procedure: Fisher's F -test

- Graphic: side-by-side box-and-whiskers plot (`boxplot(x ~ g)` or `boxplot(x1, x2)`)
- Requires: Data generated from NORMAL distribution – in *each* population
 - Requirement test: Shapiro-Wilk test
 - R function: `shapiroTest(x ~ g)`
- R function: `var.test(x ~ g)`
- R function: `var.test(x1, x2)`